



# Side effects of proton beam therapy of choroidal melanoma in dependence of the dose to the optic disc and the irradiated length of the optic nerve

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## Abstract

**Purpose** To analyze the impact of the dose to the optic disc and the irradiated length of the optic nerve on radiation-induced optic neuropathy, radiation-induced retinopathy, iris neovascularization, secondary glaucoma, enucleation, and local tumor control after proton beam therapy (PBT) of choroidal melanoma.

**Method** Retrospective analysis of 1129 patients, who received primary PBT for the treatment of choroidal melanoma with a dose of 60 cobalt gray equivalents (CGE) between 1998 and 2013 at the Helmholtz-Zentrum Berlin, Germany. Kaplan-Meier curves and logrank test have been used for time-to-event analyses. Adjustment for potential confounders was done using multiple Cox regression models with forward and backward selection.

**Results** We found a significant correlation between the irradiated length of the optic nerve and the dose to the optic disc (correlation coefficient, 0.93). Multivariate Cox regression revealed the dose to the optic disc as an independent predictive risk factor for the development of radiation-induced optic neuropathy ( $p < 0.001$ , HR 1.023, 95% CI 1.016–1.029), iris neovascularization ( $p < 0.001$ , HR 1.013, 95% CI 1.008–1.019), secondary glaucoma ( $p < 0.001$ , HR 1.017, 95% CI: 1.011–1.023) and enucleation ( $p < 0.001$ , HR 1.037, 95% CI 1.020–1.053). The irradiated length of the optic nerve was not a statistically independent predictive risk factor in multivariate analysis.

**Conclusion** Our data implicate the predominance of the dose to the optic disc over the irradiated length of the optic nerve regarding radiation-induced optic neuropathy, iris neovascularization, secondary glaucoma, and enucleation.

**Keywords** Choroidal melanoma · Proton beam therapy · Irradiated nerve length · Dose to the optic disc · Secondary glaucoma

## Introduction

In recent years, several large studies have established the role of proton beam therapy (PBT) in the treatment of uveal melanoma, which combines high local control rates with long-term eye preservation and acceptable toxicity [1–4]. Other teletherapeutic options are SRS (stereotactic radiosurgery) with Gamma Knife® and CyberKnife where a single dose is applied and SRT (stereotactic radiotherapy), a fractionated treatment with photons [5–10]. A major benefit of teletherapeutic approaches is the independence of tumor size and location. However, it is possible to irradiate para-, juxta-, and circumpapillary tumors using (slotted) plaques [10, 11]. The most common sources for brachytherapy are ruthenium-106 (beta radiation), iodine-125 (gamma radiation), and palladium-103 (gamma radiation) [10–12].

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Despite of it all, if the tumor is situated on or in the direct vicinity of the optic disc and/or the macula, at least a partial irradiation of these sensitive structures is inevitable [2]. Irradiation-induced side effects such as neuropathy, retinopathy, and reduced visual acuity have consistently been correlated with posterior tumor location independent of type of irradiation [2, 6, 11]. In these cases, some authors argue that radioactive plaque therapy or stereotactic radiotherapy (e.g., Gamma Knife® and CyberKnife) may produce comparable visual outcome, local control rates, and eye retention rates at lower costs and better availability.

The optic nerve with its anterior part, the optic disc, and the fovea are seen as the most radiosensitive structures in posterior uveal melanoma patients. A complete sparing of the optic disc is frequently impossible, especially if the tumor is abutting the disc. In contrast, the dose to the episcleral nerve can reliably be reduced taking full advantage of the steep dose gradient of PBT. The question remains if sparing of the optic nerve leads to a reduction of irradiation-induced side effects due to the serial nature of the optic disc and optic nerve.

Therefore, the objective of this study is to analyze the impact of the dose on the disc depending on the irradiated nerve length.

## Material and methods

This study was approved by the institutional review board of the Charité - Universitätsmedizin Berlin, Berlin, Germany, and was conducted in accordance with the tenets of the Declaration of Helsinki.

In total, 2327 patients were irradiated for uveal melanoma between 1998 and 2013. In this study, we included 1129 patients who received primary PBT for the treatment of choroidal melanoma with 60 cobalt gray equivalents (CGE; 1 CGE = 1.1 Gray (Gy)). We excluded all patients with ciliary body or iris involvement as well as those patients with adjuvant tumor resection or salvage PBT for local recurrence.

Before irradiation, four tantalum clips were sutured to the sclera. Ocular ultrasound was performed in order to obtain tumor dimensions and to verify clip-to-clip, clip-to-tumor, and tumor-to-disc distances. Tumor margins were identified within wide-angle fundus photography. Based on these data and additional high-resolution CT scan of the eye, clip and tumor geometrical information were merged into a coherent virtual eye model. All treatment plans were generated using the EYEPLAN software until 2006 [13] and OCTOPUS afterwards [14] aiming to minimize the dose to the optic disc, episcleral nerve, fovea, lens, and ciliary body without compromising 100% coverage of the clinical target volume (CTV). Clinical target volume and gross tumor volume (GTV) are identical. In all patients, an additional margin was added to the CTV to comprise the planning target volume (PTV). In

total, a distal safety margin of 2.5 mm was used: One millimeter was added to account for patient movement and an additional 1.5 mm was added to the depth in beam direction to account for inaccuracies of axial eye length and deviations in tissue density.

We finally optimized our treatment plans in order to keep the dose to the optic disc and episcleral nerve as low as possible, even in cases where the tumor was surrounding the optic disc. In 945 patients (83.7%), we used a single wedge to take full advantage of the steep distal dose gradient sparing the posterior part of the optic nerve. In selected cases ( $n = 57$ ; 5.0%), a double wedge was applied for distal dose cut-off: one for a reduction of the dose to the optic nerve and one for a reduction of the dose to the macula. The remaining 127 (= 11.2%) patients did not need a wedge. The mean length of the optic nerve irradiated and the dose to the optic disc was determined from the dose volume-length-histogram (EYEPLAN) or the dose-volume-histogram (OCTOPUS) calculated by the treatment planning software.

PBT was performed on four consecutive days in fractions with a single dose of 15 CGE and a total dose of 60 CGE as described before [2, 4].

In yearly follow-up intervals, patients underwent a detailed ophthalmological examination including fundoscopy, ultrasound examination, and visual acuity assessment. The presence of radiation-induced optic neuropathy was defined as any sign of edema, pallor, and hemorrhage of the optic disc or optic nerve fiber layer infarction. Radiation-induced retinopathy was diagnosed clinically by fundoscopy and was defined as a new onset of retinal capillary changes including microaneurysms or dilation of retinal vessels, cotton wool spots, hemorrhages, and exudates [4]. Secondary glaucoma was defined as persisting intraocular pressure of more than 20 mmHg in patients without known history of glaucoma of another origin or < 20 mmHg, if the patients received anti-glaucomatous medication for intraocular pressure regulation.

Statistical analysis was performed using SPSS Version 25. Descriptive analyses included absolute and relative frequencies for categorical variables or mean, standard deviation, and range for quantitative measurements. Accordingly, crossover group comparisons were performed using chi-square or Mann-Whitney  $U$  test. Spearman's correlation coefficient was used for association between quantitative variables. Kaplan-Meier curves and logrank test have been used for time-to-event analyses, presenting with corresponding figures and mean times to events with 95% confidence intervals (95% CI). Adjustment for potential confounders was using multiple Cox regression models with forward and backward selection. According effects are presented as median survival and hazard ratio (HR) with 95% confidence interval. Statistical significance was assumed for  $p$  values  $\leq 0.05$ . Due to the explorative character of the investigation, no Bonferroni correction was performed.

## Results

### Basic results

We analyzed the oncological outcome and side effects of PBT for the treatment of choroidal melanoma in 1129 patients treated between 1998 and 2013. Five hundred eighty-five (51.8%) of the patients were male and 544 (48.2%) were female.

The mean age at the time of PBT was 61 years (range 16–88 years). The mean tumor thickness was  $4.1 \pm 2.0$  mm (range 1.2–14.2 mm). The mean tumor diameter was  $11.8 \pm 3.5$  mm (range 3.0–23.4 mm). According to the TNM Staging System classification of the American Joint Committee on Cancer, 8th edition [15], the tumors were categorized as: T1a: 37.6% ( $n = 425$ ), T1c: 0.3% ( $n = 3$ ), T2a: 44.5% ( $n = 502$ ), T2c: 0.6% ( $n = 7$ ), T3a: 14.3% ( $n = 162$ ), T3c: 0.5% ( $n = 6$ ); and T4a: 2.1% ( $n = 24$ ). The mean tumor-fovea distance was  $1.3 \pm 1.8$  mm (range 0.0–12.4 mm). The mean tumor-optic disc distance was  $2.0 \pm 2.1$  mm (range 0.0–13.4 mm).

The mean follow-up was  $63 \pm 37$  months. A total of 1032 patients completed the first-year follow-up. After 2 years, 935 patients were still receiving follow-ups, reducing to 769 after 3 years, 654 after 4 years, and 489 patients after 5 years. In Kaplan-Meier analysis, overall-survival was 94.9%, metastasis-free survival was 87.6%, and local tumor control rate was 96.7% after 5 years. The median time until the detection of local recurrence was 18 months (mean time,  $25 \pm 16$  months, range 3–89 months). Enucleation-free survival was 96.3% after 5 years. In total, 36 enucleations had to be performed during the follow-up period, of which 13 (36.1%) were performed for documented or suspected local recurrence. The median time until enucleation was 29 months (mean time  $32 \pm 21$  months, range 3–81 months). Iris neovascularization developed after a median time of 24 months (mean time  $29 \pm 14.5$  months, range 2–134 months). Secondary glaucoma was diagnosed after a median time of 24 months (mean time  $31 \pm 18.7$  months, range 1–142 months).

Radiation-induced optic neuropathy developed after a median time of 20 months (mean time  $24 \pm 13.5$  months, range 1–126 months) and radiation-induced retinopathy after a median time of 19 months (mean time  $22 \pm 11.3$  months, range 1–100 months).

In Kaplan-Meier analysis, glaucoma-free survival was 86.1% after 3 years and 82.2% after 5 years, respectively. In Kaplan-Meier analysis, radiation-induced optic neuropathy-free survival was 50.9% after 3 years and 40.9% after 5 years, respectively. Kaplan-Meier estimators for the development of secondary glaucoma and radiation-induced optic neuropathy in dependency of the dose to the optic disc and irradiated length of the optic nerve are shown in Figs. 1, 2, 3, and 4.

Median visual acuity (VA) before PBT was 0.4 logMAR (range – 0.1 until 2.1 logMAR,  $n = 1129$  patients), 0.5 logMAR (range – 0.1 until 2.2 logMAR,  $n = 1031$  patients)

during the first year and declined to 1.1 logMAR in the fifth year (range – 0.1 until 2.2 logMAR,  $n = 488$  patients). In total, 12.5% of the patients treated with PBT for choroidal melanoma had a VA of 1.0 logMAR or worse before PBT. The Kaplan-Meier estimators for the absence of visual impairment  $\geq 1.0$  logMAR was 32.0%/5 years.

### Correlation of dependent variables

We found a significant correlation between the irradiated nerve length and dose to the optic disc (correlation coefficient: 0.93). There was no significant correlation between dose to the fovea and dose to the ciliary body itself or with regards to the dose to the optic disc or the irradiated nerve length.

### Univariate and multivariate analyses

We performed uni- and multivariate analyses to evaluate the impact of the dose to the optic disc and the irradiated length of the optic nerve in an overall comparison to other known clinical risk factors. We considered the following risk factors: age, sex, diabetes mellitus, arterial hypertension, dose to the lens, dose to the ciliary body, dose to fovea, tumor-fovea distance, and tumor-optic disc distance. The analyses were performed for the development of radiation-induced optic neuropathy, radiation-induced retinopathy, iris neovascularization, secondary glaucoma, enucleation, and local recurrence.

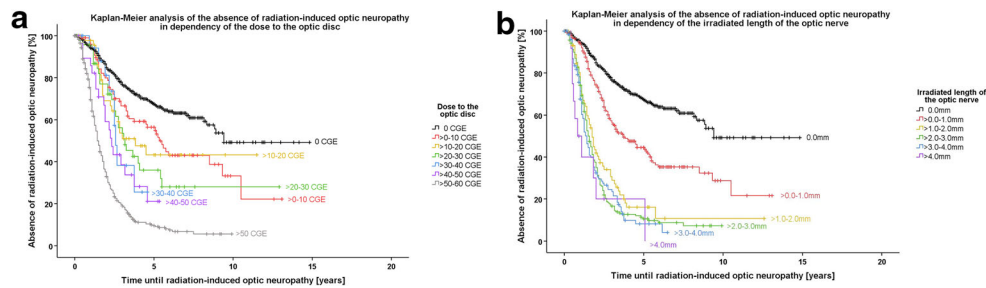
### Time-to-event analyses

There was no statistically significant difference in the logrank (Mantel-Cox) regarding the time until the development of radiation-induced optic neuropathy, radiation-induced retinopathy, secondary glaucoma, enucleation, or local recurrence for sex, diabetes, or arterial hypertension. There was a statistically significant difference between the time until the development of iris neovascularization for diabetes mellitus ( $p < 0.001$ ), but not for sex or arterial hypertension.

### Cox regression

In multivariate Cox regression after forward and backward regression, a higher dose to the optic disc was found to be an independent risk factor for the development of radiation-induced optic neuropathy ( $p < 0.001$ , HR 1.023, 95% CI 1.016–1.029), iris neovascularization ( $p < 0.001$ , HR 1.013, 95% CI 1.008–1.019), secondary glaucoma ( $p < 0.001$ , HR 1.017, 95% CI 1.011–1.023) and enucleation ( $p < 0.001$ , HR 1.037, 95% CI 1.020–1.053). The dose to the optic disc was not a significant independent risk factor for local recurrence or radiation-induced retinopathy.

Although the irradiated length of the optic nerve was a significant risk factor for the development of radiation-



**Fig. 1** Kaplan-Meier analysis of the absence of radiation-induced optic neuropathy in dependency of the **a** dose to the optic disc and **b** irradiated length of the optic nerve. Kaplan-Meier analysis shows a rather homogenous dependence of the dose to the optic disc (Fig. 1a), while there

induced optic neuropathy ( $p < 0.001$ , HR 1.661, 95% CI 1.572–1.755), radiation-induced retinopathy ( $p < 0.001$ , HR 1.202, 95% CI 1.145–1.262), iris neovascularization ( $p < 0.001$ , HR 1.362, 95% CI 1.243–1.493), secondary glaucoma ( $p < 0.001$ , HR 1.526, 95% CI 1.375–1.692) and enucleation ( $p < 0.001$ , HR 1.684, 95% CI 1.334–3.126) in univariate analysis, it was not in multivariate analysis.

An overview regarding the impact of other risk factors (e.g., tumor size and location) on the different variables is presented in Table 1 for univariate Cox regression and in Table 2 for multivariate Cox regression after forward and backward regression.

## Subgroup analysis

We performed a subgroup analysis for radiation-induced optic neuropathy and secondary glaucoma in relation to the dose to the optic disc and the irradiated length of the optic nerve. The subgroups for the dose to the optic disc were the following: 0

seems to be a threshold with no further increase of the risk for the absence of radiation-induced optic neuropathy for the irradiation of the optic nerve  $> 1.0$  mm (Fig. 1b)

CGE,  $> 0–10$  CGE,  $> 10–20$  CGE,  $> 20–30$  CGE,  $> 30–40$  CGE,  $> 40–50$  CGE, and  $> 50–60$  CGE. The subgroups for the irradiated length of the optic nerve were 0 mm,  $> 0–1.0$  mm,  $> 1.0$  mm,  $> 2.0$  mm,  $> 3.0$  mm, and  $> 4.0$  mm.

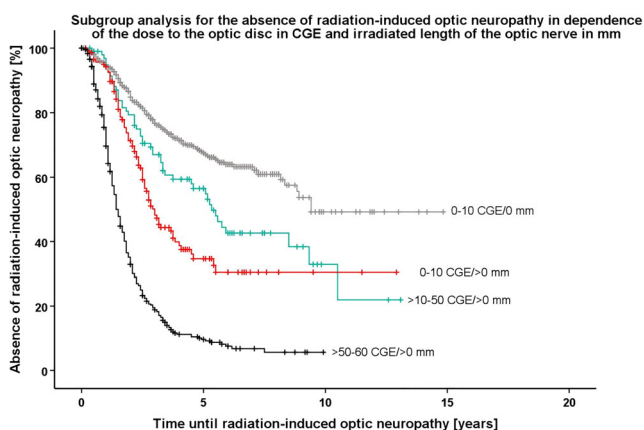
## Subgroup analysis for radiation-induced optic neuropathy

Kaplan-Meier analysis revealed a rather homogenous dependency of the dose to the optic disc for the absence of radiation-induced optic neuropathy (logrank test/pairwise comparison, 0 CGE and  $> 50$  CGE vs. all other subgroups,  $p \leq 0.01$ ). The differences between the subgroups with a dose  $> 0$  CGE up to  $\leq 50$  CGE were statistically insignificant (Fig. 1a). The hazard ratio for 0 CGE vs.  $> 0$  until  $\leq 50$  CGE was 2.066 (95% CI 1.621–2.633,  $p < 0.001$ ) and 2.456 (95% CI 2.215–2.724,  $p < 0.001$ ) for 0 CGE vs.  $> 50$  CGE.

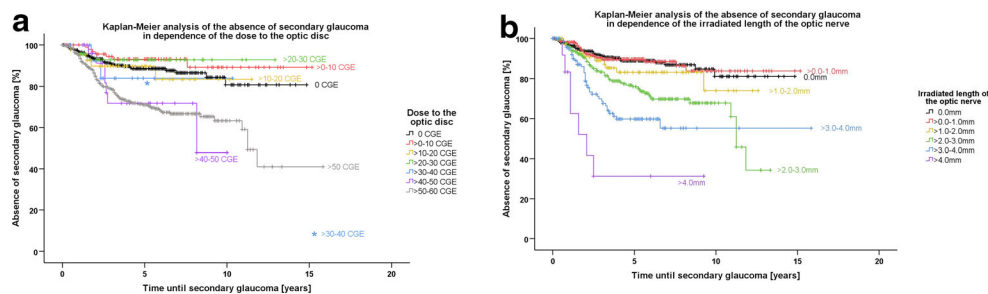
In comparison, there seems to be a threshold with no further significant increase of the risk for the absence of radiation-induced optic neuropathy for an irradiated length of the optic nerve  $> 1.0$  mm (logrank test/pairwise comparison: 0 mm vs. all other subgroups and  $> 0–1.0$  mm vs. all other subgroups,  $p < 0.001$ ). The differences between the subgroups  $> 1.0$  mm were insignificant (Fig. 1b). The hazard ratio for 0 mm vs.  $> 0–1.0$  mm was 2.089 (95% CI 1.631–2.675,  $p < 0.001$ ) and 2.379 (95% CI 2.148–2.635,  $p < 0.001$ ) for 0 mm vs.  $> 1.0$  mm.

To further analyze the impact of the dose to the optic disc and the irradiated length of the optic nerve, the following subgroups were chosen:

Low (0–10 CGE,  $n = 577$ ), intermediate ( $> 10–50$  CGE,  $n = 141$ ), and high ( $> 50$  CGE,  $n = 411$ ) dose to the optic disc with or without irradiation of the optic nerve (Figs. 2 and 4). The group with a dose to the optic disc from 0 to 10 CGE without irradiation of the optic nerve contained 478 patients. There were only seven patients with 0.6 CGE and one patient with 2.4 CGE in this subgroup. Ninety-nine patients received a dose of 0–10 CGE to the optic disc (mean dose  $3.95 \pm 2.8$  CGE, median 3.64 CGE, range 0–9.6 CGE) and had a mean irradiated length of the optic nerve of  $0.2 \pm 0.2$  mm (median



**Fig. 2** Subgroup analysis for the absence of radiation-induced optic neuropathy in patients treated with proton beam therapy for choroidal melanoma in dependence of the dose to the optic disc in CGE and irradiated length of the optic nerve in mm. This figure demonstrates the increasing risk for radiation-induced optic neuropathy in patients with an irradiation of the optic disc and the optic nerve in comparison to those patients with a low dose to the optic disc and no irradiation of the optic nerve (grey line)



**Fig. 3** Kaplan-Meier analysis of the absence of secondary glaucoma in dependence of the **a** dose to the optic disc and **b** irradiated length of the optic nerve. Pairwise comparison in the logrank test revealed a significant difference between glaucoma-free survival of patients with a dose to the optic disc  $\leq 40$  CGE and those  $> 40$  CGE ( $p < 0.05$ ) (Fig. 3a). The

difference in the logrank test for the Kaplan-Meier estimators for patients without irradiation of the optic nerve and those with an irradiated length of the optic nerve  $\leq 2.0$  mm were statistically insignificant. The risk significantly escalated in patients with an irradiated nerve length  $> 2.0$  mm (logrank test  $p < 0.001$ ) (Fig. 3b)

0.1 mm, range 0.1–1.4 mm). The optic nerve was (partially) irradiated in all patients of the intermediate dose group (median 24.6 CGE, mean  $27.0 \pm 11.4$  CGE). The mean irradiated length of the optic nerve was  $0.6 \pm 0.5$  mm (median 0.44 mm, range 0.1–3.6 mm). In the high dose group, the mean dose was  $59.2 \pm 1.8$  CGE (median 60 CGE, range 54.6–60 CGE) and the mean irradiated length of the optic nerve was  $2.8 \pm 0.7$  mm (median 2.8 mm, range 1.0–6.0 mm).

Except for four patients with no dose to the optic disc and an irradiated length of the optic nerve of only 0.1 mm, there were no patients with an irradiation of the optic nerve without simultaneous irradiation of the optic disc. A dose to the optic disc as low as 0.12 CGE already resulted in an additional irradiation of the optic nerve.

The difference between the groups with a low, intermediate, and high dose to the optic disc in relation to radiation of the optic nerve for the absence of radiation-induced optic retinopathy was significant (logrank/pairwise comparison: dose to the optic disc  $\leq 10$  CGE without irradiation of the optic

nerve vs. all other groups  $p \leq 0.002$ ; Fig. 2). In comparison to the patients with a dose to the optic disc  $\leq 10$  CGE and no irradiation of the optic nerve, the risk for the absence of radiation-induced optic neuropathy significantly increased with an additional irradiation of the optic nerve and increasing dose to the optic disc ( $\leq 10$  CGE: HR 1.650, 95% CI 1.190–2.286,  $p = 0.003$ ;  $\leq 50$  CGE: HR 1.351, 95% CI 1.230–1.484,  $p < 0.001$ ;  $> 50$  CGE: HR 1.432, 95% CI 1.375–1.493,  $p < 0.001$ ).

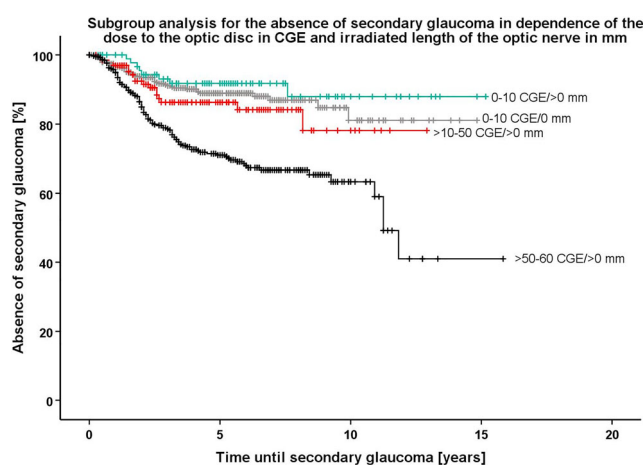
### Subgroup analysis for secondary glaucoma

Pairwise comparison in the logrank test revealed a significant difference between glaucoma-free survival of patients with a dose to the optic disc  $\leq 40$  CGE and those with  $> 40$ –60 CGE ( $p < 0.05$ ) (Fig. 3a). The hazard ratio was 3.027, 95% CI 2.222–4.125 (Cox regression: dose to the optic disc  $\leq 40$  CGE vs.  $> 40$ –60 CGE).

The differences in the logrank test for the Kaplan-Meier estimators for patients without irradiation of the optic nerve and those  $\leq 2.0$  mm were statistically insignificant (Fig. 3b). They were highly significant for patients with an irradiated nerve length  $> 2.0$  mm in comparison to those without irradiation of the optic nerve (all  $p < 0.001$ ).

The risk for a secondary glaucoma significantly escalated in patients with an irradiated nerve length  $> 2.0$  mm. The hazard ratio for patients without irradiation of the optic nerve vs. an irradiated length of the optic nerve was 1.352 (95% CI 1.183–1.545,  $p < 0.001$ ) for  $> 2.0$ –3.0 mm, 1.412 (95% CI 1.267–1.574,  $p < 0.001$ ) for  $> 3.0$ –4.0 mm and 1.606 (95% CI 1.378–1.876,  $p < 0.001$ ) for  $> 4.0$  mm after Cox regression.

The difference between the groups with a low, intermediate, and high dose to the optic disc in dependence of the irradiated length of the optic nerve was significant for the group with a dose  $> 50$  CGE in comparison to all other groups (logrank/pairwise comparison:  $p \leq 0.002$ ; Fig. 4). All other constellations were insignificant. The HR for the absence of secondary glaucoma was 1.236 (95% CI 1.152–1.326,  $p <$



**Fig. 4** Subgroup analysis for the absence of secondary glaucoma in dependence of the dose to the optic disc and the irradiated length of the optic nerve. The risk for the absence of secondary glaucoma was significantly higher in patients with a dose to the optic disc  $> 50$  CGE (logrank/pairwise comparison,  $p \leq 0.002$ ). All other constellations were statistically insignificant

**Table 1** Univariate Cox regression

|                                                | <i>p</i> value | Hazard ratio | 95% confidence interval | Number of patients included for analysis | Number of events |
|------------------------------------------------|----------------|--------------|-------------------------|------------------------------------------|------------------|
| Results for radiation-induced optic neuropathy |                |              |                         |                                          |                  |
| Age                                            | 0.870          |              |                         | 1115                                     | 598              |
| Maximum tumor height                           | < 0.001        | 1.126        | 1.083–1.171             | 1115                                     | 598              |
| Largest tumor diameter                         | < 0.001        | 1.050        | 1.027–1.073             | 1115                                     | 598              |
| Tumor-fovea distance                           | 0.003          | 0.927        | 0.883–0.974             | 1114                                     | 597              |
| Tumor-optic disc distance                      | < 0.001        | 0.644        | 0.609–0.682             | 1114                                     | 597              |
| Dose to ciliary body                           | < 0.001        | 1.054        | 1.043–1.066             | 1114                                     | 597              |
| Dose to lens                                   | < 0.001        | 1.028        | 1.022–1.035             | 1092                                     | 581              |
| Dose to fovea                                  | 0.001          | 1.005        | 1.002–1.008             | 1115                                     | 598              |
| Dose to optic disc                             | < 0.001        | 1.030        | 1.027–1.033             | 1115                                     | 598              |
| Irradiated length of the optic nerve           | < 0.001        | 1.661        | 1.572–1.755             | 1115                                     | 598              |
| Results for radiation-induced retinopathy      |                |              |                         |                                          |                  |
| Age                                            | 0.001          | 0.992        | 0.987–0.997             | 1093                                     | 837              |
| Maximum tumor height                           | < 0.001        | 1.074        | 1.039–1.111             | 1093                                     | 837              |
| Largest tumor diameter                         | 0.092          |              |                         | 1093                                     | 837              |
| Tumor-fovea distance                           | < 0.001        | 0.930        | 0.893–0.968             | 1092                                     | 836              |
| Tumor-optic disc distance                      | < 0.001        | 0.882        | 0.852–0.913             | 1092                                     | 836              |
| Dose to ciliary body                           | < 0.001        | 1.029        | 1.019–1.040             | 1092                                     | 836              |
| Dose to lens                                   | < 0.001        | 1.016        | 1.010–1.022             | 1072                                     | 821              |
| Dose to fovea                                  | 0.007          | 1.003        | 1.001–1.006             | 1093                                     | 837              |
| Dose to optic disc                             | < 0.001        | 1.010        | 1.007–1.012             | 1093                                     | 837              |
| Irradiated length of the optic nerve           | < 0.001        | 1.202        | 1.145–1.262             | 1093                                     | 837              |
| Results for iris neovascularization            |                |              |                         |                                          |                  |
| Age                                            | 0.002          | 1.017        | 1.007–1.028             | 1122                                     | 220              |
| Maximum tumor height                           | < 0.001        | 1.322        | 1.252–1.397             | 1122                                     | 220              |
| Largest tumor diameter                         | < 0.001        | 1.077        | 1.049–1.104             | 1122                                     | 220              |
| Tumor-fovea distance                           | 0.336          |              |                         | 1121                                     | 219              |
| Tumor-optic disc distance                      | < 0.001        | 0.823        | 0.762–0.889             | 1121                                     | 219              |
| Dose to ciliary body                           | < 0.001        | 1.102        | 1.087–1.117             | 1121                                     | 219              |
| Dose to lens                                   | < 0.001        | 1.051        | 1.042–1.060             | 1098                                     | 214              |
| Dose to fovea                                  | 0.251          |              |                         | 1122                                     | 220              |
| Dose to optic disc                             | < 0.001        | 1.016        | 1.011–1.021             | 1122                                     | 220              |
| Irradiated length of the optic nerve           | < 0.001        | 1.362        | 1.243–1.493             | 1122                                     | 220              |
| Results for secondary glaucoma                 |                |              |                         |                                          |                  |
| Age                                            | 0.111          |              |                         | 1054                                     | 177              |
| Maximum tumor height                           | < 0.001        | 1.335        | 1.262–1.411             | 1054                                     | 177              |
| Largest tumor diameter                         | < 0.001        | 1.092        | 1.066–1.120             | 1054                                     | 177              |
| Tumor-fovea distance                           | 0.648          |              |                         | 1053                                     | 177              |
| Tumor-optic disc distance                      | < 0.001        | 0.798        | 0.730–0.873             | 1053                                     | 177              |
| Dose to ciliary body                           | < 0.001        | 1.100        | 1.083–1.116             | 1053                                     | 177              |
| Dose to lens                                   | < 0.001        | 1.059        | 1.049–1.068             | 1031                                     | 169              |
| Dose to fovea                                  | 0.686          |              |                         | 1054                                     | 177              |
| Dose to optic disc                             | < 0.001        | 1.020        | 1.014–1.026             | 1054                                     | 177              |
| Irradiated length of the optic nerve           | < 0.001        | 1.526        | 1.375–1.692             | 1054                                     | 177              |
| Results for enucleation                        |                |              |                         |                                          |                  |
| Age                                            | 0.613          |              |                         | 1119                                     | 36               |
| Maximum tumor height                           | < 0.001        | 1.331        | 1.192–1.486             | 1119                                     | 36               |
| Largest tumor diameter                         | 0.004          | 1.089        | 1.028–1.153             | 1119                                     | 36               |

**Table 1** (continued)

|                                      | <i>p</i> value | Hazard ratio | 95% confidence interval | Number of patients included for analysis | Number of events |
|--------------------------------------|----------------|--------------|-------------------------|------------------------------------------|------------------|
| Tumor-fovea distance                 | 0.288          |              |                         | 1118                                     | 35               |
| Tumor-optic disc distance            | 0.001          | 0.652        | 0.505–0.842             | 1118                                     | 35               |
| Dose to ciliary body                 | < 0.001        | 1.067        | 1.032–1.102             | 1118                                     | 35               |
| Dose to lens                         | 0.019          | 1.029        | 1.005–1.053             | 1095                                     | 34               |
| Dose to fovea                        | 0.169          |              |                         | 1119                                     | 36               |
| Dose to optic disc                   | < 0.001        | 1.035        | 1.019–1.051             | 1119                                     | 36               |
| Irradiated length of the optic nerve | < 0.001        | 1.684        | 1.334–2.126             | 1119                                     | 36               |
| Results for local recurrence         |                |              |                         |                                          |                  |
| Age                                  | 0.028          | 1.033        | 1.003–1.063             | 1119                                     | 34               |
| Maximum tumor height                 | 0.155          |              |                         | 1119                                     | 34               |
| Largest tumor diameter               | 0.391          |              |                         | 1119                                     | 34               |
| Tumor-fovea distance                 | 0.070          |              |                         | 1118                                     | 34               |
| Tumor-optic disc distance            | 0.391          |              |                         | 1118                                     | 34               |
| Dose to ciliary body                 | 0.081          |              |                         | 1118                                     | 34               |
| Dose to lens                         | 0.718          |              |                         | 1095                                     | 32               |
| Dose to fovea                        | 0.009          | 1.022        | 1.006–1.039             | 1119                                     | 34               |
| Dose to optic disc                   | 0.657          |              |                         | 1119                                     | 34               |
| Irradiated length of the optic nerve | 0.549          |              |                         | 1119                                     | 34               |

0.001) for patients of the low-dose group in comparison to the high-dose group.

## Discussion

Due to the anatomical location of the optic disc and the optic nerve, it seems inevitable that irradiation of the optic disc leads to subsequent damage of the optic nerve. The close relationship between these structures is reflected by the high correlation coefficient of 0.93 found in our analyses.

We investigated the contribution of the dose to the optic disc and irradiated length of the optic nerve to major treatment complications after PBT by multivariate statistical analysis. The results may allow conclusions to be drawn for different structure-sparing strategies during the treatment planning process. Both the irradiated length of the optic nerve and the dose to the optic disc were shown to be significant predictors for the development of radiation-induced optic neuropathy, radiation-induced retinopathy, secondary glaucoma, iris neovascularization, and enucleation in univariate analyses. However, multivariate analysis only revealed the dose to the optic disc, not the irradiated length of the optic nerve as a significant and independent predictive risk factor for radiation-induced optic neuropathy, iris neovascularization, secondary glaucoma, and enucleation. Therefore, despite the close correlation between the dose to the optic disc and the optic nerve, a clear predominance of the dose to the optic disc

for the development of radiation-induced effects can be assumed.

A dose-volume effect depending on length of optic nerve irradiated was previously described by Mishra et al. for developing neovascular glaucoma, one of the most common forms of secondary glaucoma after PBT [3]. The authors observed a significantly lower rate of neovascular glaucoma in cases where the optic nerve was irradiated  $\leq 1$  mm. An earlier analysis of our group also showed the irradiated length of the optic nerve to be one of the significantly independent risk factors in multivariate analysis for the development of neovascular glaucoma [16].

A reason for this might be the location of the central vein, which runs through the optic nerve for about 10 mm. In the planning software, the diameter of the optic disc and the optic nerve is given with 1.5 mm. One must be careful with the interpretation of the variable “irradiated length of the optic nerve” as it does not explicitly reflect which parts of the optic nerve have been irradiated. It must be assumed that radiation has a significantly different impact on the radiation-induced side effects if the optic nerve is irradiated 1 mm in cross-section or tangential. Nevertheless, the probability of a significant dose to the central vein increases with the irradiated length of the optic nerve. This might explain the increasing risk for the development of secondary glaucoma when the optic nerve was irradiated  $> 2.0$  mm in our series.

Comparisons of the dose-dependence of other methods are limited. In most studies, the dose to the optic disc and the

**Table 2** Multivariate Cox regression (after forward and backward regression)

|                                                | <i>p</i> value | Hazard ratio | 95% confidence interval |
|------------------------------------------------|----------------|--------------|-------------------------|
| Results for radiation-induced optic neuropathy |                |              |                         |
| Maximum tumor height                           | <0.001         | 1.117        | 1.062–1.176             |
| Largest tumor diameter                         | 0.033          | 1.032        | 1.003–1.062             |
| Tumor-optic disc distance                      | 0.007          | 0.880        | 0.802–0.965             |
| Dose to fovea                                  | 0.001          | 1.006        | 1.002–1.009             |
| Dose to optic disc                             | < 0.001        | 1.023        | 1.016–1.029             |
| Results for radiation-induced retinopathy      |                |              |                         |
| Age                                            | 0.042          | 0.995        | 0.989–1.000             |
| Maximum tumor height                           | < 0.001        | 1.086        | 1.049–1.124             |
| Tumor-fovea distance                           | < 0.001        | 0.921        | 0.883–0.960             |
| Results for iris neovascularization            |                |              |                         |
| Diabetes mellitus                              | 0.015          | 1.555        | 1.091–2.216             |
| Age                                            | 0.001          | 1.020        | 1.008–1.032             |
| Maximum tumor height                           | < 0.001        | 1.169        | 1.088–1.257             |
| Dose to ciliary body                           | < 0.001        | 1.071        | 1.049–1.092             |
| Dose to optic disc                             | < 0.001        | 1.013        | 1.008–1.019             |
| Results for secondary glaucoma                 |                |              |                         |
| Maximum tumor height                           | < 0.001        | 1.187        | 1.094–1.289             |
| Largest tumor diameter                         | 0.022          | 1.055        | 1.008–1.104             |
| Dose to lens                                   | < 0.001        | 1.031        | 1.019–1.043             |
| Dose to optic disc                             | < 0.001        | 1.017        | 1.011–1.023             |
| Results for enucleation                        |                |              |                         |
| Maximum tumor height                           | < 0.001        | 1.376        | 1.216–1.556             |
| Dose to optic disc                             | < 0.001        | 1.037        | 1.020–1.053             |
| Results for local recurrence                   |                |              |                         |
| Age                                            | 0.016          | 1.037        | 1.007–1.068             |
| Dose to fovea                                  | 0.006          | 1.023        | 1.007–1.040             |

Table 2 shows the independent risk factors for radiation-induced optic neuropathy, retinopathy, iris neovascularization, secondary glaucoma, enucleation, and local recurrence after multivariate Cox regression of the presumed predictors from the univariate Cox regression analyses presented in Table 1. For reasons of clarity, non-significant risk factors are not presented

irradiated length of the optic nerve are not further specified. Above all, it should be considered that the indication for ruthenium plaques, as the most common form of treatment of uveal melanoma, is largely complementary to PBT and other teletherapeutic approaches [7]. This is associated with a completely different risk profile of these methods with frequently higher doses to adjacent organs at risk in PBT. PBT is the preferred treatment for tumors at the posterior pole and/or tumors > 5 mm. According to the manual's instructions, ruthenium plaques are recommended for tumors < 5 mm (Eckert & Ziegler BEBIG GmbH, Berlin, Germany, "Ru-106 Eye Applicators-Instructions for Use," VersionTD04\_002/Rev. 15/05.2019). They can also be used in tumors exceeding 5 mm, taking the higher scleral dose into consideration [12]. Alternatively, iodine or palladium plaques can be used for tumors > 5 mm. In general, it is possible to irradiate tumors abutting the optic disc with slotted plaques [10, 11].

Maheshwari et al. reported a median dose to the optic nerve from 94.3 Gy (range 18.3–343.5 Gy) in peri- and parapapillary uveal melanomas treated with palladium-103 slotted plaques [11]. There is no further detailed reporting of the dose to the optic disc or the irradiated length of the optic nerve. The tumor size was comparable to our collective (mean tumor height  $5.2 \pm 2.7$  mm, mean tumor diameter  $10.7 \pm 3.4$  mm).

Due to the parapapillary location the glaucoma-free probability (89%/18 months) and the radiation-induced optic neuropathy-free probability (41%/24 months) was more unfavorable than in our collective. Besides of all, high local tumor control rates up to 98.1% may be found even in this delicate location in specialized centers, e.g., using slotted palladium-103 plaques [11]. However, the treatment success of plaque therapy, especially regarding local tumor control, depends more on the accuracy of the surgeon than in teletherapeutic approaches.

It is important to mention that for dose calculation in brachytherapy, especially with ruthenium-106, only the central depth dose curve of the plaque and the decay of ruthenium-106 is used. Reporting the dose to critical structures like macula or optic disc is not common. The reason is the absence of a CE/FDA-certified treatment planning system. To our knowledge, only a non-approved plaque simulator is available [17].

Due to the higher flexibility of the CyberKnife, the relevance of the Gamma Knife® for the treatment of uveal melanoma has progressively diminished. Data on the experience with robotic radiosurgery for the treatment of uveal melanoma are sparse. Siedlecki et al. compared their experience with robotic radiosurgery (single dose of 20 Gy administered with CyberKnife) vs. ruthenium-106 brachytherapy [7]. The treatment indications for robotic radiosurgery are similar to those for PBT, which is also represented in the significantly different tumor prominences (median tumor prominence: robotic radiosurgery 6.4 mm, ranging from 1.6 to 15.8 mm vs. ruthenium-106 brachytherapy: 4.1 mm ranging from 1.8 to 6.1 mm). They found a significantly different 5-year incidence of secondary glaucoma of 46.7% for patients treated with robotic radiosurgery and 11.1% for brachytherapy ( $p < 0.001$ ). This difference vanished in a subgroup analysis of the patients with tumors  $\leq 6$  mm (3-year incidence robotic radiosurgery: 13.8% vs. brachytherapy 11.2%,  $p = 0.138$ ).

Unfortunately, dose values to the organs at risk are not published on a regular basis. Since the dose to the tumor is significantly lower with only 20 Gy in robotic radiosurgery than in PBT, this consequently results in a lower dose to the optic nerve and optic disc.

In patients treated with (fractionated) stereotactic radiation therapy administered with CyberKnife using a dose of 10–66 Gy in 1–3 fractions and a maximum dose to the ipsilateral optic nerve with a median of 19.58 Gy (range 1.86–74.26 Gy), radiation optic neuropathy was described in 11% of the patients after median 17 months [8]. Detailed dose information for the optic nerve and the optic disc were not reported separately. However, the major disadvantage of robotic radiosurgery (5-year local tumor control rate, 70.8% [9]) and (fractionated) stereotactic radiation therapy (5-year local recurrence-free survival, 75 [7]–82% [5]), appears to be the worse local tumor control rate compared to brachytherapy (up to 98.1% [11]) and PBT (local tumor control rate at 5 years in this paper, 96.7%). A reason for the higher local recurrence rates might be the lower dose of only 20 Gy applied with CyberKnife, when used as SRS. A general issue with SRS and SRT is the fact that the irradiation is solely CT/MRI-based. Therefore, the domain of SRS and SRT are tumors with a size recognizable in CT. The treatment of flat tumors or tumor parts with  $< 1$  mm of height, not detectable on CT or MRI, is only possible to a limited extent. This might explain the higher local recurrence rates [18]. Flat tumor parts are

integrated into the planning procedure of PBT by marking the CTV during an operation with tantalum clips and on a high-resolution fundus photo of the patient, which can be directly used in the treatment planning system for CTV definition. Technically, it would be possible to optimize the CTV using tantalum clips and in SRS and SRT as well. This effort would of course increase the costs of treatment with SRS and SRT—one of the main arguments for the use of SRS and SRT besides the better availability. Due to their spherical projection, fundus images could not be used for direct CTV delineation in SRS/SRT treatment planning systems. Using plaques, it is possible to check the correct location including flat tumor parts via fundoscopy. However, the supposed benefit of the better availability of plaques should also be observed with skepticism. Most publications report the results of specialized ophthalmologic centers, which have developed great experience in the treatment of uveal melanoma. It is known from other diseases that treatment in specialized centers improves patient's outcomes [19]. Therefore, the use of plaques but also SRS and SRT in non-specialized centers warrants critical attention since it can be assumed that rates for local recurrence and side effects might be higher in these cases, especially with such a rare disease like uveal melanoma.

The data of our univariate analyses shows that the irradiated length of the optic nerve indeed has some impact, but it seems to be subordinate to the dose to the optic disc according to multivariate analysis. This raises the question of whether one could reduce the side effects by reducing the dose to the optic disc or whether this would only elevate the risk of local recurrence.

A dose of 60–70 CGE delivered in 4–5 fractions is considered a standard dose in most centers performing PBT for the treatment of choroidal melanoma [20]. The available data on normal tissue tolerance after irradiation of the optic nerve in general predicts a high risk of radiation-induced optic neuropathy when using typical doses and fractionation schemes for radiotherapy. There is evidence that radiation tolerance is increased with a reduction in the dose per fraction [21]. Radiation-induced optic neuropathy and retinopathy are also described after PBT at considerably lower doses, e.g., 20 CGE in patients with choroidal hemangioma [22]. Although incidences were much lower, it documents that there are no clear cut-off threshold doses for the development of these side effects.

However, our data indicate that partial sparing of the optic nerve or the optic disc during irradiation results in a significantly lower risk for the development of radiation-induced optic neuropathy and glaucoma. This might be one reason for the high rate of posttreatment globe preservation in our patient cohort.

Our results regarding the overall (94.9% at 5 years) and metastasis-free survival (87.6% at 5 years) are largely in line

with available long-term data after PBT of uveal melanoma [23]. Our long-term eye retention rate was excellent with 96.3% after 5 years. This may be partially due to the absence of ciliary body involvement in our patients. The high eye retention rate of about 95.2% without compromising overall and disease-free survival supports continuing the use of PBT in this cohort.

This study has some limitations that are not just related to its retrospective design. During the treatment preparation process, the eye and its structures are integrated into a virtual model in order to generate the treatment plan. While most intraocular structures can easily be fitted into this model, the optic nerve is of variable appearance. These variations are not always accurately reflected by the treatment planning software.

In conclusion, our data suggest that the dose to the optic disc is the predominating risk factor for the development of radiation-induced optic neuropathy, iris neovascularization, secondary glaucoma, and enucleation. With proton beam therapy, a maximum sparing of the optic disc is possible and has major impact on the avoidance of optic neuropathy and secondary glaucoma in these patients.

## Compliance with ethical standards

**Conflict of interest** A.I. Riechardt: reimbursements of travel expense: EURETINA 2015/Novartis. A. Stroux, I. Seibel, J. Heufelder, O. Zeitz, D. Böhmer, A. M. Jousen, and J. Gollrad state that there are no conflicts of interest.

**Ethical approval** All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. For this type of study, formal consent is not required. This article does not contain any studies with animals performed by any of the authors.

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